

2 (a)	I is proportional to V i.e. $\frac{I}{V} = \text{constant}$ From Fig. 2.1 $V = 6.0 \text{ V}$, $I = 1.25 \text{ A}$, $\frac{I}{V} = 0.208$ $V = 8.4 \text{ V}$, $I = 1.75 \text{ A}$, $\frac{I}{V} = 0.208$ $V = 10.8 \text{ V}$, $I = 2.25 \text{ A}$, $\frac{I}{V} = 0.208$ (b) (i) Since the ratio $\frac{I}{V} = 0.208$ is constant, I is proportional to V (shown) (ii) Resistance $= \frac{V}{I} = \frac{8.4}{1.75} = 4.8 \Omega$ (iii) - Let N be negative terminal of battery Current in wire AB $= \frac{V}{R_{AN}} = \frac{12}{4.0 + 5.0} = 1.333 \text{ A} \approx 1.3 \text{ A}$ - Current in resistor X $= \frac{V}{R_{X+Y}} = \frac{12}{4.8 + 2.7} = 1.6 \text{ A}$ Let N be negative terminal of battery. $V_{CN} = \frac{R_{CN}}{R_{AN}} \times E = \frac{1.0 + 5.0}{4.0 + 5.0} \times 12 = 8.0 \text{ V}$ $V_{DN} = \frac{R_Y}{R_{X+Y}} \times E = \frac{2.7}{4.8 + 2.7} \times 12 = 4.32 \text{ V}$
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	Potential difference between C and D = $V_{CN} - V_{DN} = 8.0 - 4.32 = 3.68 \text{ V}$
3 (a)	The loss in the electromagnetic radiation's individual photon energy must be greater than the work function of the metal. The difference between the loss in the